Worksheet 5D: Trigonometric Identities and Equations

What This Worksheet Is About

Trigonometric identities are equations that are true for all values of the variable. They're not problems to solve — they're tools that let us simplify expressions, prove relationships, and solve equations.

These identities appear constantly in physics, engineering, and signal processing. Mastering them gives you powerful algebraic tools.

Part A: Fundamental Identities (18 problems)

The Pythagorean Identity

From the unit circle, we know:

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

Derived forms: - $\sin^2\theta = 1 - \cos^2\theta$ - $\cos^2\theta = 1 - \sin^2\theta$

Simplify using the Pythagorean identity:

1. $\sin^2(x) + \cos^2(x)$

2. $1 - \sin^2(x)$

3. $1 - \cos^2(x)$

4. $\frac{\sin^2(x)}{1-\cos^2(x)}$

5. $\frac{1-\sin^2(x)}{\cos(x)}$

6. $(\sin(x) + \cos(x))^2$

Hint: Expand, then use the identity.

Quotient and Reciprocal Identities

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad \cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \qquad \csc\theta = \frac{1}{\sin\theta}$$

Simplify:

7. $\frac{\sin(x)}{\cos(x)}$

8. $\sin(x) \cdot \sec(x)$

9. $\cos(x) \cdot \csc(x)$

10. $\tan(x) \cdot \cos(x)$

11. $\frac{\sec(x)}{\csc(x)}$

12. $\sin^2(x) + \sin^2(x)\tan^2(x)$

Hint: Factor out $\sin^2(x)$.

More Pythagorean Identities

Dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$:

$$\boxed{\tan^2\theta + 1 = \sec^2\theta}$$

Dividing by $\sin^2\theta$:

$$\boxed{1 + \cot^2\theta = \csc^2\theta}$$

Simplify:

13. $\sec^2(x) - 1$

14. $\csc^2(x) - 1$

15. $\sec^2(x) - \tan^2(x)$

16. $\frac{\tan^2(x)}{\sec^2(x)-1}$

17. $\tan^2(x) + \cot^2(x) + 2$

Hint: Rewrite using $\sec^2 - 1$ *and* $\csc^2 - 1$.

18. $\frac{1-\cos^2(x)}{\sin(x)}$

Part B: Sum and Difference Formulas (20 problems)

The Formulas

$$\boxed{\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B}$$

$$\boxed{\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B}$$

$$\boxed{\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}}$$

Memory aid: - Sine: "sine-cosine, cosine-sine" with SAME sign as the original - Cosine: "cosine-cosine, sine-sine" with OPPOSITE sign

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Find exact values using sum/difference formulas:

19. $\sin(75°) = \sin(45° + 30°)$

20. $\cos(75°)$

21. $\sin(15°) = \sin(45° - 30°)$

22. $\cos(15°)$

23. $\tan(75°)$

24. $\sin(105°)$ — *Write as 60° + 45°*

25. $\cos(105°)$

Verify identities:

26. Show that $\sin(x + \frac{\pi}{2}) = \cos(x)$

27. Show that $\cos(x - \frac{\pi}{2}) = \sin(x)$

28. Show that $\sin(\pi - x) = \sin(x)$

29. Show that $\cos(\pi - x) = -\cos(x)$

** Science Connection: Phase in Wave Addition**

30. When two waves of the same frequency but different phases combine:

$$y_1 = A\sin(\omega t) \quad \text{and} \quad y_2 = A\sin(\omega t + \phi)$$

Using the sum formula on y_2:

$y_2 = A[\sin(\omega t)\cos\phi + \cos(\omega t)\sin\phi]$

$y_1 + y_2 = A\sin(\omega t) + A\sin(\omega t)\cos\phi + A\cos(\omega t)\sin\phi$

$= A(1 + \cos\phi)\sin(\omega t) + A\sin\phi\cos(\omega t)$

If $\phi = \frac{\pi}{2}$, simplify this expression.

31. If $\phi = \pi$ (waves completely out of phase), what is $y_1 + y_2$?

** Science Connection: Rotation Matrices**

32. In computer graphics and robotics, rotation by angle θ uses the matrix:

$$R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

To rotate first by α, then by β, you multiply matrices. The result should be rotation by $\alpha + \beta$. This is equivalent to showing:

$\cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) = \cos(\alpha + \beta)$

Verify this matches the sum formula for cosine!

33. Rotating a point $(1, 0)$ by 30°, then by 60°, should give the same result as rotating by 90°. - (a) What is $\cos(90°)$ and $\sin(90°)$? - (b) Verify that $\cos(30°)\cos(60°) - \sin(30°)\sin(60°) = \cos(90°)$

** Astronomy: Orbital Mechanics**

34. The position of a planet in an elliptical orbit involves angles that add and subtract.

If a satellite's angular position is the sum of two angles: $\theta_{total} = \theta_1 + \theta_2$

where $\theta_1 = 45°$ and $\theta_2 = 60°$, find $\cos(\theta_{total})$ and $\sin(\theta_{total})$ exactly.

Part C: Double Angle and Half Angle Formulas (18 problems)

Double Angle Formulas

Setting $A = B$ in the sum formulas:

$$\boxed{\sin(2\theta) = 2\sin\theta\cos\theta}$$

$$\boxed{\cos(2\theta) = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta}$$

$$\boxed{\tan(2\theta) = \frac{2\tan\theta}{1 - \tan^2\theta}}$$

Simplify:

35. $2\sin(30°)\cos(30°)$

36. $\cos^2(45°) - \sin^2(45°)$

37. $2\cos^2(30°) - 1$

38. $1 - 2\sin^2(60°)$

Find exact values:

39. If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I, find: - (a) $\cos\theta$ - (b) $\sin(2\theta)$ - (c) $\cos(2\theta)$

40. If $\cos\theta = \frac{5}{13}$ and θ is in Quadrant I, find $\sin(2\theta)$ and $\cos(2\theta)$.

Half Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

(Sign depends on which quadrant $\frac{\theta}{2}$ is in)

Find exact values:

41. $\sin(22.5°)$ — *Use* $\theta = 45°$

42. $\cos(22.5°)$

43. $\sin(15°)$ — *Use* $\theta = 30°$ — Compare to your answer from Problem 21!

44. $\cos(15°)$

** Science Connection: Power Reduction**

In electronics, it's often necessary to convert between powers of trig functions and multiple angles.

45. The power reduction formulas come from solving the double-angle formulas:

$$\sin^2\theta = \frac{1-\cos(2\theta)}{2} \qquad \cos^2\theta = \frac{1+\cos(2\theta)}{2}$$

Verify by expanding $\cos(2\theta) = 1 - 2\sin^2\theta$ and solving for $\sin^2\theta$.

46. In AC circuits, power is proportional to $\sin^2(\omega t)$ or $\cos^2(\omega t)$.

Using the power reduction formula, show that: $\sin^2(\omega t) = \frac{1}{2} - \frac{1}{2}\cos(2\omega t)$

What does this tell you about the frequency of power oscillation compared to voltage oscillation?

** Science Connection: Interference Patterns**

47. When two identical beams of light interfere, the intensity is proportional to:

$$I \propto \cos^2\left(\frac{\phi}{2}\right)$$

where ϕ is the phase difference.

Using the half-angle relationship, rewrite this as: $I \propto \frac{1+\cos\phi}{2}$

48. In Young's double-slit experiment, the phase difference is $\phi = \frac{2\pi d \sin\theta}{\lambda}$ where d is slit separation, θ is angle from center, and λ is wavelength.

At what values of ϕ do you get: - (a) Maximum intensity (bright fringes)? - (b) Minimum intensity (dark fringes)?

** Science Connection: Lithography Interference**

49. Interference lithography creates patterns by interfering two laser beams. The pattern has period:

$$p = \frac{\lambda}{2\sin\theta}$$

where θ is the half-angle between the beams.

Using $\lambda = 193$ nm and $\theta = 30°$, what pattern period can be achieved?

50. To create 10 nm features, you need $p = 20$ nm. If $\lambda = 13.5$ nm (EUV), what angle θ is needed?

Part D: Solving Trigonometric Equations (20 problems)

Strategy

1. Isolate the trig function if possible
2. Use identities to simplify
3. Find all solutions in the required interval
4. Remember: trig functions are periodic!

Solve for x in $[0, 2\pi)$:

51. $\sin(x) = \frac{1}{2}$

52. $\cos(x) = -\frac{\sqrt{2}}{2}$

53. $\tan(x) = 1$

54. $\sin(x) = -\frac{\sqrt{3}}{2}$

55. $2\cos(x) - 1 = 0$

56. $2\sin(x) + \sqrt{3} = 0$

Solve (may require factoring or identities):

57. $2\sin^2(x) - 1 = 0$

58. $\cos^2(x) - \cos(x) = 0$

59. $2\sin^2(x) - \sin(x) - 1 = 0$

60. $\tan^2(x) - 3 = 0$

61. $\sin(2x) = \frac{\sqrt{3}}{2}$ in $[0, 2\pi)$

Hint: Let $u = 2x$, solve for u in $[0, 4\pi)$, then find x.

62. $\cos(2x) = -\frac{1}{2}$ in $[0, 2\pi)$

Use identities to solve:

63. $\sin^2(x) - \cos^2(x) = 0$

Hint: This is $-\cos(2x) = 0$

64. $\sin(x)\cos(x) = \frac{1}{4}$

Hint: Use $\sin(2x) = 2\sin(x)\cos(x)$

65. $1 - \sin^2(x) = \frac{1}{2}$

** Science Connection: Resonance**

66. A system oscillates at frequency ω. Resonance occurs when the driving frequency matches the natural frequency, i.e., when:

$\sin(\omega_d t) = \sin(\omega_n t)$

If $\omega_d = 2\pi$ rad/s and $\omega_n = 2\pi$ rad/s, at what times t in $[0, 1)$ seconds do the phases align (equal to each other)?

** Science Connection: Standing Waves**

67. A standing wave on a string is described by:

$y(x, t) = 2A\sin(kx)\cos(\omega t)$

The nodes (points that don't move) occur where $\sin(kx) = 0$.

If $k = \frac{\pi}{0.5} = 2\pi$ m^{-1}, find all nodes in the interval $x \in [0, 2]$ meters.

** Science Connection: Diffraction Minima**

68. In single-slit diffraction, dark fringes occur when:

$\sin\theta = \frac{m\lambda}{a}$

where a is slit width, λ is wavelength, and $m = \pm 1, \pm 2, ...$

For $a = 1000$ nm and $\lambda = 500$ nm, find all angles θ where dark fringes occur (within the physical range where $|\sin\theta| \leq 1$).

General Solutions:

69. Give the general solution (all solutions) for $\sin(x) = \frac{1}{2}$.

70. Give the general solution for $\cos(x) = 0$.